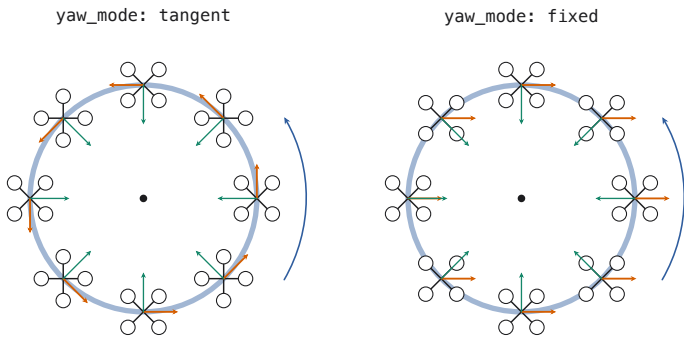
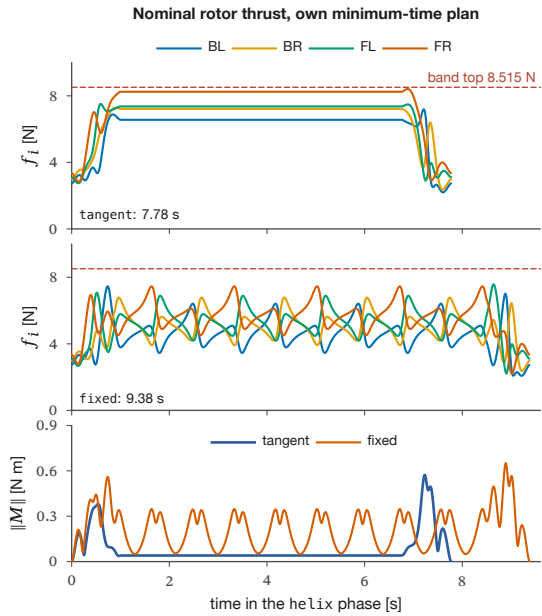


a Heading along the helix top view, 1 of 5 turns, counter-clockwise



Red: body x (heading); green: horizontal part of the thrust tilt (toward the axis).
Tangent: the axis is always on the vehicle's left, so the tilt is fixed in body axes and the body turns at a constant rate. Fixed: the axis direction sweeps once around the body per turn, so roll/pitch must oscillate, which needs periodic torques and uneven rotor thrusts.

b



c Why helix.yaml uses the tangent heading

	tangent	fixed
helix time (minimum)	7.78 s	9.38 s
cruise speed	4.70 m/s	3.76 m/s
steady-speed limit	4.79 m/s	3.92 m/s
rotor thrust in cruise	6.56–8.25 N, constant	3.45–7.46 N, oscillating
torque in cruise	≤ 0.04 N m	up to 0.35 N m
nominal tilt (max)	66°	55°
yaw rate in cruise	4.64 rad/s	0

Same geometry and limits (configs/helix.yaml, $\eta = 0.85$), synthetic plant. In tangent the rotor thrusts are constant; they differ mainly because of the plant's CoM offset. Its cost is a yaw rate of 4.64 rad/s, which is why helix.yaml raises max_yaw_rate_rad_s to 2π . In fixed the busiest rotor's oscillation plus the lag lead $\dot{f} + \tau \ddot{f}$ reaches the band top first.